

Problem 10.7

Bruce deposits 100 into a bank account. His account is credited interest at a nominal rate of interest i convertible semiannually. At the same time, Peter deposits 100 into a separate account. Peter's account is credited interest at a force of interest of δ . After 7.25 years, the value of each account is 200. Calculate $i - \delta$.

Solution

For Bruce's account, the effective interest rate per half-year is $i/2$. There are

$$2(7.25) = 14.5$$

half-year periods in 7.25 years. Therefore,

$$200 = 100 \left(1 + \frac{i}{2} \right)^{14.5}.$$

Dividing by 100 gives

$$2 = \left(1 + \frac{i}{2} \right)^{14.5}.$$

Thus,

$$1 + \frac{i}{2} = 2^{1/14.5},$$

so

$$i = 2 \left(2^{1/14.5} - 1 \right) = 0.09792851065.$$

For Peter's account,

$$200 = 100e^{7.25\delta}.$$

Therefore,

$$e^{7.25\delta} = 2,$$

and

$$\delta = \frac{\ln 2}{7.25} = 0.09560650766.$$

Hence,

$$i - \delta = 2 \left(2^{1/14.5} - 1 \right) - \frac{\ln 2}{7.25}.$$

Numerically,

$$\boxed{i - \delta = 0.002322003}.$$